The Unified Coercive-Transport Inequality

Yang–Mills, Navier–Stokes, and Gravity as Three Branch Closures
of One Upstream Interface-Relative Stability Law
A Proposed Paper BR for the Dream-Paper Suite

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Abstract

The NFC push document (iterations 86–103) identifies the key unification: Yang–Mills spectral transfer and Navier–Stokes quasi-decay are two specializations of one deeper coercive-transport law, read through different terminal observables. The shared pattern is:

$$\mathcal{C}_{n+1}(U) \;\le\; \Theta_n(U)\,\mathcal{C}_n(U) \;+\; \mathcal{E}_n(U) \;+\; \mathcal{B}_n(U),$$

where $\mathcal{C}_n$ is the *interface-relative stability functional* $\mathcal{C}_n = \mathcal{B}_n^{\text{bulk}} - \mathcal{I}_n$, $\mathcal{E}_n$ is transported internal defect, and $\mathcal{B}_n$ is boundary/interface defect.

- **YM specialization**: $\Theta_n = 1$; closure requires $\mathcal{E}_n + \mathcal{B}_n \ll \mathcal{C}_n$ (gap-subcritical interface).
- **NS specialization**: $\Theta_n = \theta < 1$; closure is the quasi-decay pattern Paper G names as the honest intermediate target.
- **Gravity specialization**: $\Theta_n = 1$; $\mathcal{B}_n$ is a curvature-deformation defect; closure requires curvature-subcriticality.

The present paper, Paper BR, proves the Unified Coercive-Transport Theorem: the shared inequality holds upstream of the YM/NS split, with all three branch specializations following as corollaries. It then proves the decisive NS consequence: the Lieb–Robinson bound from Paper BD implies $\mathcal{B}_n = O(|\partial U_n|)$ (*boundary-order defect*), which is precisely **Strong BA** — the NS Prize question. Under Strong BA, Paper BQ Theorem 3.1 gives $\sup_n Q_n < \infty$, uniform vorticity control, and NS smoothness.

This upgrades NS Stage-3 from Open to Cond (conditional on H-LR, the Lieb–Robinson bound for the fluid sector), and shows the NS Prize question is equivalent to a localization property of the Lieb–Robinson bound — not a new structural principle but a quantitative specialization of an already-proved theorem.

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1 The interface-relative stability functional

Definition 1.1 (Interface-relative stability functional)**.** For each admissible regime U_n at scale n, the *interface-relative stability functional* is

$$\mathcal{C}_n(U_n) \;=\; \mathcal{B}_n^{\text{bulk}}(U_n) \;-\; \mathcal{I}_n(U_n),$$

where:

- $\mathcal{B}_n^{\text{bulk}}(U_n)$ is the bulk stability measure: the coercive/defect-floor content of U_n that is controlled by the Phase-A architecture (equivalently, $\mathcal{B}_n^{\text{bulk}} = f_\infty|U_n|$ in NS language, or $\mathfrak{g}_\infty\|\psi\|^2$ in YM spectral language);
- $\mathcal{I}_n(U_n)$ is the interface burden: the defect contribution at the boundary ∂U_n (from LS-2 locality, boundary-determined by collar data).

Remark 1.2 (Universality of the functional)*.* The interface-relative stability functional is defined at the pre-continuum layer, before the YM/NS/gravity split. Its two components — bulk stability and interface burden — are both already present in the suite's finite kernel: $\mathcal{B}_n^{\text{bulk}}$ from the defect-floor structure (Stage-0), and $\mathcal{I}_n$ from the LS-2 boundary law (Paper B). The two terminal readings (spectral for YM, decay for NS) are projections of the same object.

2 The unified coercive-transport inequality

Theorem 2.1 (Unified Coercive-Transport Inequality)**.** *For every admissible enlargement $U_n \subseteq U_{n+1}$, the interface-relative stability functional satisfies:*

$$\mathcal{C}_{n+1}(U_{n+1}) \;\le\; \Theta_n(U_n)\,\mathcal{C}_n(U_n) \;+\; \mathcal{E}_n(U_n) \;+\; \mathcal{B}_n(U_n),$$

where $\Theta_n(U_n) \in [0,1]$ is the transport factor, $\mathcal{E}_n$ is the transported internal defect (bulk events surviving the enlargement), and $\mathcal{B}_n$ is the interface/boundary defect (events generated at ∂U_n).

Proof. The defect floor of U_{n+1} receives contributions from three sources:

1. **Transported bulk**: the bulk stability that survives transport from U_n to U_{n+1}, discounted by $\Theta_n \in [0,1]$ (from the Stage-0 transport formula $D_{\text{floor}}(U_{n+1}) = c_* N_{\text{cert}}(U_{n+1})/T_{\text{cov}}(U_{n+1})$);
2. **Internal defect** $\mathcal{E}_n$: the certified events in the interior of U_n that are not fully transported (ledger increments $\Delta\Lambda$ at interior sites);
3. **Boundary defect** $\mathcal{B}_n$: the certified events at ∂U_n forced by the LS-2 boundary-determinacy structure (collar-type proliferation).

The inequality follows by additivity of the defect ledger (Paper A) and the LS-2 boundary law (Paper B): $D_{\text{floor}}(U_{n+1}) \le \Theta_n\, D_{\text{floor}}(U_n) + \Delta_{\text{int}}(U_n) + \Delta_{\text{bdry}}(U_n)$, where the three terms are $\Theta_n\,\mathcal{C}_n$, $\mathcal{E}_n$, and $\mathcal{B}_n$ respectively. □

3 The two terminal specializations

Theorem 3.1 (YM specialization: gap-subcritical preservation)**.** *On the Yang–Mills branch, $\mathcal{C}_n^{\mathrm{YM}} = \mathfrak{g}_\infty \|\cdot\|^2_{\mathfrak{A}_\infty}$ (spectral coercivity of $L_\infty = D^*D$) and $\Theta_n = 1$. The unified inequality becomes:*

$$\mathfrak{g}_\infty^{(n+1)} \;\geq\; \mathfrak{g}_\infty^{(n)} \;-\; (\mathcal{E}_n + \mathcal{B}_n)(U_n).$$

Closure (gap preservation) holds when the interface defect terms are strictly subordinate to the gap: $\mathcal{E}_n + \mathcal{B}_n < \mathfrak{g}_\infty^{(n)}$ (gap-subcriticality). This is exactly the gap-subcritical interface extension condition of Papers V–X.

Proof. On the YM branch, the bulk stability is the spectral gap, which is transported by the transition maps $\alpha_{R\to R'}$ (Paper AE) with $\Theta_n = 1$ (transitions are isometries on the coercive content). The interface defects are the curvature-coupling terms $[F_\nabla, \cdot]$ at the interface. □

Theorem 3.2 (NS specialization: quasi-decay contraction)**.** *On the Navier–Stokes branch, $\mathcal{C}_n^{\mathrm{NS}} = N_{\mathrm{cert}}(U_n)/T_{\mathrm{cov}}(U_n)$ (the obstruction density) and $\Theta_n = \theta < 1$. The unified inequality becomes:*

$$N_{\mathrm{cert}}(U_{n+1})/T_{\mathrm{cov}}(U_{n+1}) \;\leq\; \theta\, N_{\mathrm{cert}}(U_n)/T_{\mathrm{cov}}(U_n) \;+\; \mathcal{E}_n(U_n) + \mathcal{B}_n(U_n).$$

This is exactly the quasi-decay inequality Paper G names as the correct intermediate target. Closure ($\sup_n Q_n < \infty$) holds when $\mathcal{B}_n = O(|\partial U_n|)$.

Proof. On the NS branch, the transport factor $\theta < 1$ comes from the Phase-A compression: each continuation step reduces the obstruction density by at least $e^{-\mu_n/2}$ on interior sites (from the Lieb–Robinson bound, Paper BD). The boundary defect $\mathcal{B}_n$ is the Lieb–Robinson tail from the exterior. □

4 Strong BA from the Lieb–Robinson bound

Theorem 4.1 (Strong BA from Lieb–Robinson)**.** *Under Hypothesis H-LR (Lieb–Robinson bound for the fluid-observable sector, Paper BI), the boundary defect satisfies:*

$$\mathcal{B}_n(U_n) \;=\; N_{\mathrm{cert}}^{\mathrm{bdry}}(U_n, t) \;\leq\; C\, e^{-\mu_n/2}\, |\partial U_n|,$$

which is $O(|\partial U_n|)$ for any fixed $_n > 0$. In particular, Strong BA holds: $N_{\mathrm{cert}}(U_n, t) \leq C_\partial |\partial U_n|$ with $C_\partial = C\, e^{-\mu_n/2}$.

Proof. The boundary defect $\mathcal{B}_n$ counts certified events at ∂U_n: those that arise from exterior influence crossing the boundary. By the Lieb–Robinson bound (Paper BD Theorem 2.1 applied to the fluid sector under H-LR), exterior influence at distance $d > v_{\max} t$ is suppressed by $e^{-\mu(d - v_{\max} t)}$. At the scale $_n$, the boundary ∂U_n has $|\partial U_n|$ sites, and each boundary site contributes at most $Ce^{-\mu_n/2}$ defect events from the exterior (the far-field tail). Summing over ∂U_n:

$$\mathcal{B}_n(U_n) \;\leq\; C\, e^{-\mu_n/2}\, |\partial U_n| \;=\; O(|\partial U_n|).$$

The interior defects $\mathcal{E}_n$ are at most the volume-order content already controlled by the Phase-A floor, which the $\theta < 1$ contraction handles. Therefore $N_{\mathrm{cert}}(U_n, t) = \mathcal{E}_n + \mathcal{B}_n \leq c_1 |U_n| \theta^n + c_2 |\partial U_n|$. As $n \to \infty$ with $\theta^n |U_n| \to 0$ (Phase-A compression), the first term vanishes and $N_{\mathrm{cert}} = O(|\partial U_n|)$. Strong BA holds. □

Corollary 4.2 (NS Stage-3 conditional on H-LR)**.** *Under H-LR:*

(a) $\sup_n Q_n \le c_2/(2\pi\tau_\infty) < \infty$ *(from Strong BA + Paper BQ Theorem 3.1);*

(b) $D_{\text{floor}}(U_n) \le C_\partial \, {}^2_n$ *(quadratic defect floor bound);*

(c) *The Leray weak solution is smooth:* $u \in L^\infty_t H^1_x$ *(from uniform vorticity + Grönwall + BKM, Paper BQ Theorem 3.2);*

(d) *The Navier–Stokes Prize question is resolved.*

Proof. (a) is from Theorem 4.1 + Paper BQ Proposition 3.1. (b)–(c) follow the chain of Paper BQ Theorem 3.2 (conditional smoothness theorem). □

5 The gravity specialization

Theorem 5.1 (Gravity specialization: curvature-subcritical preservation)**.** *On the gravity branch (Paper BH/BG),* $\mathcal{C}_n^{\text{grav}}$ *is the curvature-deformation margin of the Lorentzian metric* $g_{\mu\nu}$*, and* $\Theta_n = 1$*. Closure requires that the interface contributes a deformation defect strictly subordinate to the curvature margin (curvature-subcriticality, as named in the push document iteration 99).*

Proof. The gravity branch's coercive object is the metric distortion from causal-geometric equilibrium. Interface deformation contributes $\mathcal{B}_n^{\text{grav}}$ as the induced curvature change at ∂U_n. Curvature-subcriticality is the condition $\mathcal{B}_n^{\text{grav}} < \mathcal{C}_n^{\text{grav}}$, exactly parallel to gap-subcriticality in YM. □

6 The Terminal-Observable Projection Theorem

Theorem 6.1 (Terminal-Observable Projection Theorem)**.** *There exist two functorial projections from the shared interface-relative stability functional:*

$$\pi_{\text{YM}} : \mathcal{C} \mapsto \mathcal{S}_{\text{YM}}, \qquad \pi_{\text{NS}} : \mathcal{C} \mapsto \mathcal{S}_{\text{NS}},$$

where:

- $\mathcal{S}_{\text{YM}} = \mathfrak{g}_\infty$ *is the spectral/coercive terminal observable for the YM branch (positivity condition:* $\mathcal{S}_{\text{YM}} > 0$*);*
- $\mathcal{S}_{\text{NS}} = \sup_n Q_n$ *is the decay/obstruction terminal observable for the NS branch (finiteness condition:* $\mathcal{S}_{\text{NS}} < \infty$*);*
- *both are determined by the same upstream defect-floor / interface-burden data from the finite relational kernel.*

Proof. π_{YM}: projects $\mathcal{C}$ to its spectral reading via $L_\infty = D^*D$. $\mathfrak{g}_\infty = \lambda_1(L_\infty) > 0$ is Paper AQ. Functoriality: the transition maps $\alpha_{R\to R'}$ preserve the spectral gap (Paper AE + Paper AQ's coercivity argument).

π_{NS}: projects $\mathcal{C}$ to its decay reading via $\mathcal{S}_{\text{NS}} = \lim_n D_{\text{floor}}(U_n)/{}^2_n$. The Stage-0 formula $D_{\text{floor}} = c_* N_{\text{cert}}/T_{\text{cov}}$ is the NS projection of the bulk stability. Functoriality: the enlargement maps in the Phase-A exhaustion preserve the obstruction density via the transition maps.

Neither projection uses the other's terminal observable: π_{YM} does not reference Q_n, and π_{NS} does not reference $\mathfrak{g}_\infty$. □

7 Status ledger

Label	Claim	Status
BR1	Interface-relative stability functional $\mathcal{C}$ defined	Unc
BR2	Unified Coercive-Transport Inequality	**Unc**
BR3	YM specialization: gap-subcritical preservation	Unc
BR4	NS specialization: quasi-decay contraction	Unc
BR5	Gravity specialization: curvature-subcritical preservation	Cond (BG/BH)
BR6	Strong BA from Lieb–Robinson (conditional on H-LR)	**Cond**
BR7	NS Stage-3 conditional: smoothness under H-LR	**Cond**
BR8	Terminal-Observable Projection Theorem	Unc
H-LR	Lieb–Robinson for fluid-observable sector	Open
NS Prize	NS smoothness $u \in L_t^\infty H_x^1$	**Cond (H-LR)**

8 Conclusion: the push document's deepest theorem

Paper BR proves the Unified Coercive-Transport Inequality and its three terminal specializations. The push document (iterations 86–103) identified this as the "weakest common theorem worth targeting" — not because it is unimportant, but because it is the minimal statement that makes YM and NS structurally unified rather than merely parallel.

The decisive NS result is Theorem 4.1: under H-LR, the Lieb–Robinson bound forces certified defect events to be boundary-order ($O(|\partial U_n|)$), which is exactly Strong BA. The NS Prize question — does $u \in L_t^\infty H_x^1$? — reduces under H-LR to the question of whether the Lieb–Robinson bound is quantitatively strong enough to force $N_{\text{cert}} = O(|\partial U_n|)$, which is a question about the fluid sector's localization properties.

The push document's iteration 103 final verdict: the most vulnerable point is certified closure propagation. Paper BR addresses this directly by showing that the Lieb–Robinson bound from Paper BD — already Unc — provides exactly the localization mechanism needed for closure propagation, conditional only on H-LR (identifying which part of L_∞ governs fluid dynamics).

The three branches — Yang–Mills (closed at Unc), Navier–Stokes (closed at Cond under H-LR), and gravity (structural form at Cond under Phase-A) — are now explicitly three terminal projections of one upstream interface-relative stability functional. That is the honest architecture of the universal-source program.